

An object-tracking technique for counting grape clusters in Brazilian Northeast's pergola vineyards

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Abstract—In precision viticulture, deep learning and computer vision have been increasingly employed to automate grape cluster counting, a crucial task for yield estimation and farm management. However, existing methods fail to address the unique challenges posed by the pergola vine training system — a widely used but understudied method, particularly in Brazil's Northeast, where viticulture plays a key role in economic and social development. To address these limitations, we propose a novel, lightweight approach that combines object detection (YOLOv8) and tracking (ByteTrack) to count grape clusters in handheld smartphone videos captured under diverse environmental conditions. Unlike previous works, our technique is applied to the unique layout of the pergola system, where occlusions, perspective variations, and irregular cluster distributions complicate automated counting. We evaluate our method on real-world vineyard footage, achieving 75.1% mAP@50 and a 69.9% F1-score for cluster detection, with a final cluster count only 27% above the actual value—establishing the first baseline for this challenging task. By enabling cost-effective, accessible yield estimation, our work contributes to improving efficiency and equity in viticulture. We publicly release our implementation to encourage further research and practical adoption (https://github.com/artsasse/pergola-grape-count/blob/main/grape_cluster_count.ipynb).

Index Terms—grape cluster counting, pergola training system, object detection, object tracking, computer vision, deep learning

I. INTRODUCTION

The integration of computer vision and deep learning is revolutionizing various fields. In the agricultural sector, these innovations have boosted productivity by leveraging data collected through sensors and vision-based systems to optimize decision-making. The fusion of these data sources enables a more precise analysis of farm activities, integrating information from different sources to enhance monitoring and agricultural management [1].

In grape cultivation, these new technologies have been widely employed to predict yield, detect pests, automate processes with robots, map vineyard areas, among other tasks.

This study was partially supported by Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - CAPES, Conselho Nacional de Desenvolvimento Científico e Tecnológico - CNPq (grants 182318/2024-9 and 131477/2024-2) and Fundação Carlos Chagas Filho de Amparo à Pesquisa do Estado do Rio de Janeiro - FAPERJ (grant 201.467/2022).

Yield prediction, in particular, is a key aspect of precision viticulture and can involve several subtasks, such as grape cluster and berry counting, grape classification based on quality and ripeness, and crop weight estimation. Grape cluster and berry counting are essential for estimating vineyard productivity and optimizing harvest planning. Traditional methods, based on manual sampling, are labor-intensive and prone to errors, making computer vision and deep learning approaches promising alternatives. Convolutional neural networks (CNNs) [2], and multiple object trackers (MOTs) [3] have been used to identify and count fruits in images and videos, often surpassing conventional methods in accuracy and scalability [4].

In addition to the inherent inaccuracies of manual counting, the limited frequency of data collection can undermine the reliability of yield estimates, particularly given the dynamic nature of agricultural environments. For instance, factors such as early ripening, extreme weather conditions, or damage caused by pests and strong winds can lead to the premature loss of grape clusters. Without data freshness [5], farmers may rely on outdated estimates, resulting in suboptimal decisions regarding irrigation, harvesting, and commercial distribution. Automating cluster counting through efficient computational methods based on computer vision and deep learning would enable more frequent monitoring of vineyard production, allowing real-time strategic adjustments.

In this context, we propose, implement, and validate a novel technique for automatically counting grape clusters, specifically designed for the pergola vine training system. This system, predominantly used in vineyards across Brazil's South and Northeast regions [6], [7], presents a unique structural layout that introduces challenges in image analysis not addressed in previous studies. Given that grape production in Northeastern Brazil has been linked to more equitable income distribution [8], and that the pergola system demands greater labor and costs compared to alternative training methods [7], bridging this research gap represents a valuable step toward fostering social and economic benefits in the region. The main contributions of this research are:

- A technique to automatically count grape clusters in RGB videos of pergola system vineyards, a previously unex-

plored context in grape cluster detection and counting.

- A quantitative evaluation of the performance of our technique when applied to real-world data.

The document is divided as follows: in Section II we delve into the grape cluster counting problem and possible approaches; in Section III, we compare this work with others focused on grape cluster counting and detection; in Section IV, we explain the proposed technique, the experimental setup and the dataset used; in Section V we present the results of our tests, discussing our findings; in Section VI we comment on the limitations of our study and future work; and in Section VII we synthesize our findings and perspectives about grape cluster counting.

II. PROBLEM DEFINITION AND HYPOTHESES

Several computational methods for grape counting have been proposed in the literature, but many of them exhibit significant limitations. Some techniques estimate only the number of individual berries rather than entire grape clusters, which may not be suitable for certain agricultural applications. Others are designed exclusively for static images, making them unsuitable for full-video analysis, or require specialized recording equipment, limiting accessibility for many farmers. Drone-based counting methods, for example, also face challenges in vineyards with dense foliage, where cluster visibility is obstructed, as illustrated in Fig. 1.

Besides these technical challenges, none of these methods have been specifically tested in vineyards using the pergola system [9], which features an arched canopy structure, as illustrated in Fig. 2. The pergola system is the most widely used vine training method in vineyards across Brazil's South and Northeast regions [6], [7]. Unlike traditional row-based vineyards, where grape clusters form a relatively uniform "wall" at a consistent distance from the camera, pergola vineyards create a tunnel-like structure where clusters are distributed at varying depths. This results in significant variations in their apparent size and the occurrence of occlusion, due to perspective, making analysis more challenging compared to previous studies, where cameras could be positioned perpendicularly to the clusters for a more uniform view. Moreover, some of the previous techniques are only described in research papers, with no publicly available implementations, making it difficult to reproduce results and adapt them to different data.

In the studied vineyard, the grape cluster counting is currently performed manually, by workers who walk along the vineyard rows, annotating the quantity of clusters in each of them. We decided to proceed with a counting technique based on processing videos captured in a similar way, by someone traversing these vineyard rows. This way, we expect our work to be easily applied to real-world farms and extended to be used as an application in a smartphone, for example, without disrupting much of the traditional process, but ensuring a higher efficiency.



Fig. 1. Aerial image captured by drone at the vineyard where the images were captured.



Fig. 2. Frame from video captured inside one of the vineyard corridors.

A. Approach Selection

Based on Farjon et al.'s survey about counting methods in agriculture using deep learning [4], we highlight four possible approaches for this task:

- 1) Direct counting and weakly supervised methods, which estimate the number of fruits without requiring individual identification;
- 2) Object detection combined with 2D tracking, where a detector identifies fruits in each video frame, and a tracker associates these detections over time to prevent duplicate counting;
- 3) Object detection combined with 3D mapping, which reconstructs a three-dimensional scene to enhance counting accuracy;
- 4) Density estimation, which generates a heat map of the target objects in each image and try to regress the number of objects from that representation.

The density estimation approach, despite bringing a good performance and demanding less in terms of image annotation efforts, has been mainly applied to static images, especially those from Unmanned Aerial Vehicles (UAVs), which can not be used in our case, since the dense foliage at the top of the pergola vineyards prevents the capture of useful images for counting. Actually this protection against the sunlight is one

of the main reasons the pergola training system is used in the Northeast of Brazil [6].

The direct counting and weakly supervised methods suffer from a similar problem, since they are direct towards the analysis of static images, not keeping track of repeated fruits along successive frames.

The object detection methods are best suited for our case, since we are using videos instead of static images. Between the 2D and the 3D tracking approaches, we chose to implement the former since it is computationally cheaper than the latter in general [10] and we found more detailed examples of its use in counting tasks, which could serve as inspiration for our own technique. Many 3D mapping techniques also require the usage of specific hardware, like stereo cameras or depth sensors, demanding more than just RGB videos, which are our only resources in this study.

More specifically, our selected approach first employs a convolutional deep learning-based detection method to learn and predict bounding boxes for grape clusters in the frames for each captured video. Then based on the predicted bounding boxes for the successive frames, a tracking algorithm assigns identifiers to unique grape clusters, keeping track of their trajectories. The effective counting is based on the total number of identifiers assigned during the video reproduction. Even though the pergola system videos differ considerably in layout, perspective and occlusion, to the data captured in other studies, as the camera moves deeper into the vineyard, each grape cluster still define a trajectory, which can be uniquely tracked. To evaluate our technique with pergola-specific data, we collected video footage from a table grape farm located in the São Francisco Valley, a key hub for irrigated fruit production in Brazil's Northeast. The technique's implementation and evaluation and the dataset captured are described in detail in Section IV.

Following Dasarthy's characterization [11], we can identify three main fusion steps in our technique:

- 1) **Object Detection:** Data Input - Feature Output (DaI - FeO), through the feature extraction from pixels by CNNs, and Feature Input - Decision Output (FeI - DeO), through the prediction of object locations in each video frame.
- 2) **Object Tracking:** Temporal Decision Input - Decision Output (DeI - DeO), through the association of trajectories between detected objects in successive video frames.
- 3) **Object Counting:** Decision Input - Decision Output (DeI - DeO), through the processing of the raw number of tracks to compute the quantity of grape clusters.

B. Research Questions and Metrics

Since our objective is to propose a cluster counting technique adequate to pergola vineyards, we defined two research questions to guide our evaluation, one focused on each component of our method (detection and tracking):

- **RQ1:** How well does the technique detect clusters in static images?

- **RQ2:** How well does the technique count clusters in videos?

Farjon et al.'s survey [4] highlights the importance of measuring the performance of object detection and counting separately, whose metrics are used to evaluate each of our research questions, respectively. For RQ1, we use the following metrics:

- **Mean Average Precision (mAP):** since the number of grape clusters (positives) is much smaller than the number of other objects (negatives) in an image, the model accuracy would not be a balanced metric for evaluation. Precision and Recall are metrics that take this imbalance into account, since the former measures the fraction of detections that correspond to true grape clusters, and the latter measures the fraction of true grape clusters correctly detected. However, each of them only indicate the level of overestimation and underestimation respectively. Average Precision (AP) helps measuring both problems since it averages the Precision over all possible levels of Recall. Different levels of Recall are achieved by varying the confidence threshold of the detections [12]. In practice, this threshold should be optimized for each application, considering which error needs to be avoided the most (grape clusters going undetected or mistaken detections) to optimize counting, but the average of possible precisions gives a balanced perspective. mAP measures the average of APs for each class, but in our case we only have one class (grape clusters). We specifically use mAP for an Intersection over Union (IoU) threshold of 50% (mAP@50), which is a standard value [4]. IoU is defined by the fraction of I (area of intersection between the predicted bounding box and the ground-truth bounding box) in relation to U (union between the bounding boxes' areas). Whenever the IoU is above the determined threshold we consider the predicted bounding box as a correct detection. The AP (which is the mAP in our case) also directly corresponds to the area under the curve (AUC) of the recall-precision curve.
- **F1-Score:** Similarly to mAP, F1-Score also takes both Precision and Recall into account by calculating their harmonic mean [13]. The difference is that the F1-Score is calculated for an individual confidence threshold and typically, in practice, we aim at the confidence threshold which gives us the best F1-Score. We fix the confidence threshold in the default value used by YOLOv8, the tool we used for object detection, which will be described in detail in Section IV. Since different values of precision and recall can yield the same score, we also report these two metrics as a complement to the F1-Score.

For RQ2, regarding counting performance, we use the following metric:

- **Mean Absolute Error (MAE):** it is the absolute difference between the predicted number of grape clusters and the ground-truth quantity [14]. Since we only have one class, we don't need to average that error over

multiple classes. Although we are dealing with deep neural networks, we chose the absolute value instead of the squared one, since we don't use this error in the network learning process and it represents the error more intuitively.

III. RELATED WORK

Shen et al. [10] proposed a grape cluster counting approach using YOLOv5, a 2D tracking algorithm, and a counting line. However, their study was conducted in vineyards with a conventional training system, where the camera moves perpendicularly to the vine rows. Additionally, their source code and dataset were not publicly released, limiting reproducibility and direct comparison. Despite these differences, their methodology is the most similar to ours.

Santos et al. [15] introduced the Wine Grape Instance Segmentation Dataset (WGISD) and explored YOLO-based detection. They employed a 3D reconstruction technique to estimate cluster positions from motion across multiple images. While they demonstrated the integration of detection and 3D mapping in videos, their study does not quantitatively evaluate counting accuracy.

Jaramillo et al. [16] also applied a 2D tracking-by-detection approach for grape cluster counting using smartphone cameras. Their method relied on artificial illumination for nighttime image capture, and the camera was stabilized by a gimbal and mounted on an all-terrain vehicle (ATV) moving at a constant speed. Instead of YOLO, they used Faster R-CNN [17] for detection.

Ciarfuglia et al. [18] combined detection, segmentation, and both 2D and 3D tracking for grape cluster counting. They worked in a tendone vineyard, a trellis system similar to pergola, and used a semi-supervised annotation approach to improve counting accuracy.

Beyond grapes, tracking-by-detection strategies have been used for other fruits with similar growth patterns, such as passion fruit [19], [20] and tomatoes [21]. However, these studies were conducted under controlled conditions with cameras positioned perpendicularly to the crop rows.

Unlike previous works, our study specifically addresses pergola-trained vineyards, where grape clusters are distributed along a tunnel-like canopy, making conventional perpendicular imaging impractical. This results in significant variations in cluster distance and scale, increasing the complexity of detection and tracking. Furthermore, while many studies use fixed camera setups on drones or vehicles, our videos were recorded by people under variable daytime conditions, adding further real-world challenges. Our approach also stands out for its accessibility, as it relies solely on standard RGB images without the need for specialized sensors. Lastly, we ensure reproducibility by providing our code and model configurations, while leveraging state-of-the-art detection (YOLOv8) and tracking (ByteTrack) techniques to address this underexplored scenario.



Fig. 3. Side view of a vineyard row. At this angle, bunches in other rows could be confused with the one under review. Also, we lost sight of the bunches in the center of the aisle.



Fig. 4. Image from bottom to top inside one of the corridors of the vineyard. Although we captured the bunches in the center of the corridor better, we missed the bunches on the sides of the corridor.

IV. METHODOLOGY

First we present the data used for training and evaluation, details of the technique implementation and then we delve into how the technique was tested to answer the research questions.

A. Dataset

Our original dataset consisted of 445 manually annotated frames on the Roboflow platform, extracted from four videos recorded in the vineyard under study. Three of these videos were captured in the late afternoon and one in the morning. All recordings were made using the 26 mm camera of an iPhone XR, by the same research team member, ensuring some level of consistency in capture conditions.

To enhance dataset diversity and improve the model's generalization ability, we applied data augmentation techniques, including rotation, horizontal flipping, brightness and contrast adjustments, and slight saturation modifications. This process expanded the dataset to 1,069 frames. The data was split into training, validation, and test sets following a 88%, 8%, and 4% ratio, respectively.

It is important to note that the original images were captured at 1080p resolution with a 16:9 aspect ratio, but were resized



Fig. 5. Annotated resized image from one of the vineyard rows.

to 640x640 pixels, to optimize processing. An example of an annotated and resized image is shown in Fig. 5.

B. Implementation

Based on our literature review, we decided to implement a technique comprised of two components: object detection and object tracking. For the detection step, we selected YOLOv8 [22], which had not yet been explored in previous grape cluster counting studies. The model was trained and assessed using manually annotated frames extracted from video footage collected in the vineyard. For the tracking step, we employed ByteTrack [23], an out-of-the-box tracking algorithm that builds upon the widely used SORT [24] and DeepSORT [25].

YOLOv8 was selected for this project due to its advanced architecture, which provides high accuracy, speed, and robustness in complex scenarios, with even greater inference efficiency compared to previous versions [22]. YOLO [26] was the first deep learning one-stage detector, being characterized by its speed, with each version bringing new improvements in accuracy and efficiency [12]. One of the key advantages of YOLOv8 specifically is its enhanced feature pyramid, a mechanism that leverages multiple levels of abstraction during processing. This improvement enables the model to detect objects of various sizes with greater accuracy, which is essential for counting grape clusters that can vary significantly in scale and position within the video. Additionally, this approach minimizes information loss and enhances the model's ability to handle variations in lighting, occlusions, and perspective changes. In this study, we specifically used YOLOv8s, an optimized version that maintains detection quality while significantly reducing model size. This choice was driven by the intent to run inference on low-power edge devices, ensuring a balance between performance and computational efficiency for real-world field applications. We use the detection-oriented version of YOLOv8s, which was pre-trained on the MS-COCO dataset [27] to speed up convergence of the model.

ByteTrack [23] is a multi-object tracking tool that employs the Kalman Filter [28] and the Hungarian Algorithm [29] to associate detected objects and predicted trajectories in a video. Its unique feature is how it leverages low-probability detections, instead of focusing only on high-probability ones, like previous techniques. First, high-confidence detections are assigned to existing trajectories, ensuring accurate matching. Then, low-confidence detections are considered and linked to previously unmatched trajectories, preserving object tracks that might otherwise be discarded due to temporary detection failures. This strategy significantly improves tracking robustness and continuity, particularly in challenging scenarios where occlusion and object appearance variations impact detection accuracy. We chose ByteTrack because it offers competitive performance on Multiple Object Tracking (MOT) tasks, while keeping state-of-the-art or similar efficiency, measured by frames per second (FPS), in many of these tasks, even against more recent techniques [30]–[32]. Moreover, ByteTrack can be used as an out-of-the-box general purpose tool through the Supervision¹ library, making it easy to apply to new contexts, without needing to adapt code from specific experiments or benchmarks.

Our development pipeline was based on the Roboflow tutorial² on object counting in videos. In addition to the official YOLOv8 version provided by Ultralytics, we extensively used the Roboflow Supervision library, which streamlined both the integration with ByteTrack and the generation of annotated videos with object counts.

C. Experiment

In this study, our focus is to propose a grape cluster counting technique based on object detection and tracking and to evaluate its performance on data collected from vineyards that follow the pergola training system. To synthesize our objective, we define the following research questions:

- RQ1: How well does the technique detect clusters in static images?
- RQ2: How well does the technique count clusters in videos?

To answer these research questions, we will rely on metrics discussed in the survey by [4]:

- For object detection performance (RQ1), we will use Mean Average Precision (mAP) and F1-Score. Especially we use mAP@50, which evaluates detection performance at a 50% Intersection over Union (IoU) threshold. The IoU measures the overlap between the predicted bounding boxes and the ground truth, and mAP@50 specifically considers detections where IoU exceeds 0.5. This ensures that we account for both localization accuracy and detection confidence in our evaluation. The precision is averaged over the precision values for all confidence levels in the detections. The F1-Score is computed for the

¹<https://supervision.roboflow.com/latest/>

²<https://blog.roboflow.com/yolov8-tracking-and-counting/>

YOLOv8s predetermined confidence threshold, ensuring a balanced evaluation of precision and recall.

- For grape cluster counting accuracy (RQ2), we adopt Mean Absolute Error (MAE). Unlike squared error metrics, MAE is sufficient in this case since we are not using this error for training any neural network, but rather to assess the overall system performance in the counting task.

We trained YOLOv8s for 25 epochs, using the default hyperparameters provided by the package. We also trained the same model again for 75 epochs, a YOLOv5s [33] model for 200 epochs leveraging the pre-trained weights from another study [34] and a YOLOv12n model [35] for 200 epochs. However, all of them yielded results slightly worse than the ones obtained in our first training trial, which will be detailed in Section V. This early convergence for different models and a preliminary exploration with public but unofficial datasets available in the Roboflow platform suggest the need of more data to achieve better results. Similarly, we adopted the default settings for ByteTrack.

For the line-based counting method, we arbitrarily defined the position of the counting line based on the heatmap of detections in the training images, as shown in Fig. 6. This is certainly a hyperparameter that can be optimized in future studies to improve accuracy and robustness.

The detection stage was evaluated using a validation set of 88 images. The counting stage was assessed using a 5-second video from a separate recording, which was not part of the training set. The ground-truth cluster count was determined through detailed video annotation using the CVAT³ software.

Due to the high cost of training and annotation, only one execution of the experiment was performed. However, future studies should incorporate multiple repetitions to ensure the statistical significance of the results. The experiment was conducted in an environment with a dedicated GPU (NVIDIA GeForce RTX 4060).

V. RESULTS

Regarding RQ1 (detection performance), our technique achieved the following results for the evaluated metrics:

- **Precision:** 71%
- **Recall:** 68.9%
- **F1-Score:** 69.9%
- **mAP@50:** 75.1%

A direct comparison with previous works is not possible, as these studies focused on vineyard systems and camera perspectives very different from ours. Still, we expected to face a challenging scenario for detection due to the image perspective, which causes grape clusters in the background to appear smaller, increasing the likelihood of missed detections. Additionally, in our dataset, dense foliage, combined with perspective variations and the presence of people in the videos, contributes to partial occlusions of grape clusters, further complicating the detection task.

³<https://www.cvat.ai>

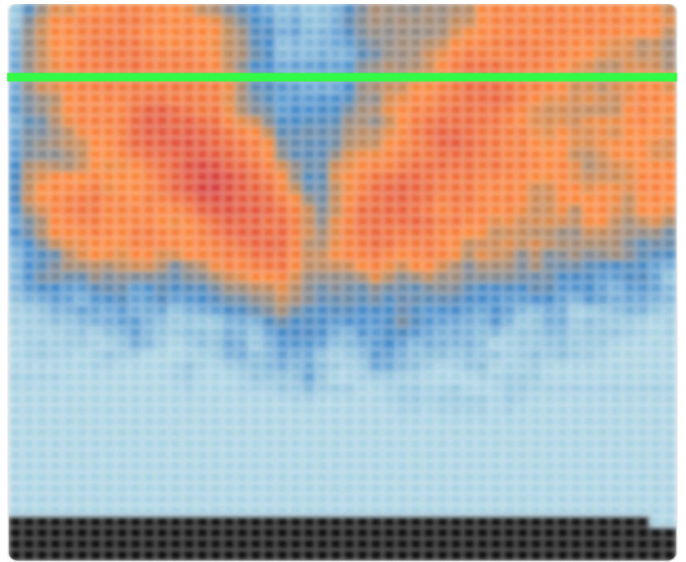


Fig. 6. Object Detection Heatmap and the counting line. With the line counting method, grape clusters would only be counted when their respective bounding boxes crossed the line.

Regarding RQ2 (counting performance), our technique, which employs two different counting methods, achieved the following results for the 5-second video extracted from unseen data, which contained 74 clusters:

- **MAE (counting by unique IDs):** 20 (27% above the actual value).
- **MAE (counting using a predefined line):** 53 (71.62% below the actual value).

Relying solely on the number of unique identifiers associated with tracks leads to an overestimation of grape clusters, indicating the need for a fusion mechanism to improve accuracy. On the other hand, the counting line—a technique used in previous works [10] to prevent double counting—results in a severe underestimation, as many grape clusters never cross the line due to the vineyard’s perspective. This highlights the necessity of a more adaptive counting strategy that accounts for the spatial distribution of clusters within the video. We hypothesize that using multiple counting lines and merging their results could yield a more precise estimate. For now, despite its overestimation, counting unique identifiers appears to be the most practical approach given our specific camera position.

Fig. 7 illustrates the algorithm in action, showcasing both the assignment of unique IDs to clusters and the use of the counting line simultaneously.

VI. LIMITATIONS AND FUTURE WORK

Regarding data, we have additional videos available from the studied farm, which could be used to further train and evaluate the proposed technique. Additionally, we could leverage publicly available datasets, such as the one by Santos et al. (2020) [15], to enhance detection performance.



Fig. 7. The proposed technique at work in one of the videos used for testing the counting part.

Concerning the models, we could explore more thoroughly the hyperparameter optimization for both detection and counting models. Also some studies incorporate auxiliary mechanisms such as attention [36] or pruning [10], which could be tested in combination with our approach.

For better counting evaluation, it would be ideal to have more videos with reliable ground-truth counts and to find a more precise method for manual counting. We could leverage the expertise of field workers who perform manual grape cluster counting as part of their daily tasks.

As stated in [3], multi-source sensor data fusion is the next frontier for tracking techniques, which could also be explored in our case. Since the videos were captured manually in an irregular terrain there are non-uniform movements of the camera which could be taken into account to avoid track association mistakes. A related problem for YOLO-based detection in drone-captured aerial images has been addressed in [37]. In other fusion level, in [38], the authors estimate 3D positions from 2D RGB images to enhance pedestrian tracking, which could be adapted to our case. Finally, with the presence of multiple sensors in modern smartphones, we could use the same equipment and resources to employ stereo or RGB-D imaging, which has already been done in other contexts [39].

VII. CONCLUSION

In this study, we proposed and evaluated a grape cluster counting technique for videos of vineyards using the pergola training system. Our method is based on modern object detection and tracking techniques and was trained and evaluated on video data collected from a grape farm in the Brazilian Semi-Arid region.

As expected, the pergola system layout, the environmental conditions and the video perspective created a challenging context for detection and counting. Nonetheless, our technique achieved significant results for both tasks, representing the first result of its kind for pergola system vineyards.

Finally, this study represents a promising step toward the application of computer vision techniques, particularly object detection and tracking methods, for automated grape cluster counting in vineyards where the pergola system is predominant

— both in the Brazilian Northeast and South regions and in other parts of the world.

ACKNOWLEDGMENT

We thank Scinet Institute, Netcon and Fazenda IPFittipaldi for the partnership and access to the vineyard.

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